

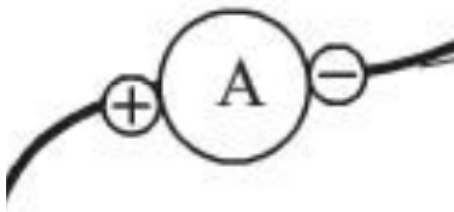
1) Elettrotecnica (EC) (salute)

1.1) Introduction

The **electric current** is defined $i(t) = \frac{dq}{dt}$. The sign of the current depends on how the surface is oriented. In 2018 the unit of measure of charge, Coulomb $[C]$ is defined as $[A \cdot s]$.

The **voltage** is defined as $\int_A^B E dt$. The result depends on the path between A, B . A positive quantity means the work is done by the field, otherwise it goes against such forces. The **positive pole** (+) corresponds with the start of the path. The unit of measure is Volt $[V] = [J/C]$. This force correlates with the **electric potential** since $V_{AB} = U_A - U_B$. We define $U = \frac{dW}{dq}$ With $U = 0$ at infinite distance

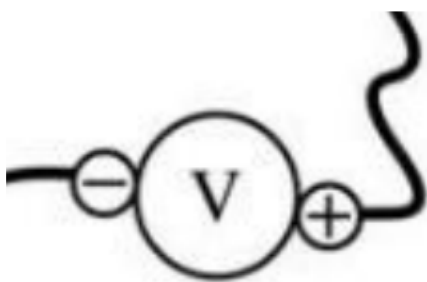
1.2) Measuring Devices



Ammeter example

This is an ammeter, it uses the **load convention**. It measures the current passing through this bipole/branch. An ideal ammeter doesn't alter the current.

In AC regime, it only measures the effective value that is always positive



Voltmeter example

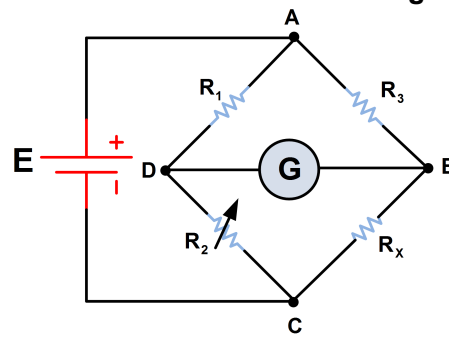
This is a voltmeter, it measures the difference of potential on its two poles. An ideal Voltmeter has infinite resistance in order to create an open circuit.

In AC regime, it only measures the effective value that is always positive

There are two ways to measure the voltage and current in a circuit:

- V-I Resistance measurement with Voltmeter directly in parallel with the Resistance: This causes a systematic error in the Ammeter by adding the internal resistance of the ammeter
- V-I Resistance measurement with Ammeter directly in series with the Resistance: This causes a systematic error in the Voltmeter by adding the resistance of the Ammeter

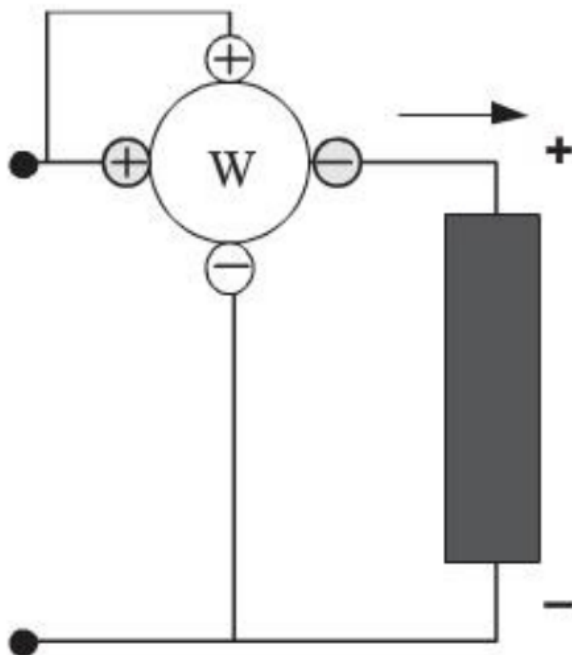
There is also a way to measure a resistance: **The Wheatstone bridge**



Wheatstone bridge

R_2 is changed until the Ammeter (Galvanometer) measures 0 current, then:

$V_{CD} = V_{CB}$ and also $V_{AD} = V_{AB}$, then $R_X I_X = R_2 I_2$ and $R_1 I_1 = R_3 I_3$, but since $I_1 = I_2$ and $I_X = I_3$ we can divide both sides



Wattmeter example

In order to measure the power a wattmeter is used. This device is essentially a voltmeter and an ammeter combined.

In AC regime it only measures the mean power, i.e. the real/active power

$$P = VI \cos(\phi)$$

1.3) Electric Network Model

A network is a model that can describe electromagnetic phenomena under the following conditions:

- It is possible to define integral parameters of the scalar and vector fields
- We can assume an infinite speed of propagation
Moreover it must also be in **quasi stationary regime**. this means:
- No radiation
- Small dimension s.t. the propagation is instantaneous

We base our network on **bipoles** connected together via **conductors**. We define the following fundamental conditions:

- The conductors are ideal
- The space outside of network components is insulated (quasi stationary)

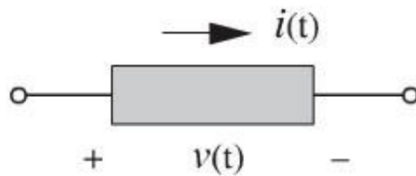
- The field is conservative outside of bipoles, i.e. a conductor has no difference of potential
- No charge is accumulated outside of bipoles

We mainly study bipoles, but generally we can define n-poles. Let's focus on bipoles:

A bipole must follow these two fundamental conditions:

- The current entering is the same as the one exiting
 - Between the two terminals there is a difference of potential
- These are just a recap of the fundamental conditions described above, in particular the second one states that outside of the bipole the field is conservative.

On a bipole we also define a convention based on how the current should enter:



Load convention in a generic bipole

1. Load convention: I is positive if it enters on + side
2. Source convention: I is negative if it enters on + side

There are various categories of bipoles:

- Active bipole: $I = 0$, $V_{AB} \neq 0$
- Passive bipole: I and V_{AB} have the same sign
 - it is always an inert bipole: $I=0$, $V=0$
- Linear/Non-linear: self explanatory

1.4) Power

An important parameter to define is power $p(t) = v_{AB}(t)i(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta Q}{\Delta T}$. Based on the convention used for a bipole, we can better understand the sign of this parameter: A *source convention(generator)* will have a positive power if it is delivering power, while a *load convention(resistor)* will have a positive power when it is absorbing power.

THEOREM 1 (ENERGY CONSERVATION PRINCIPLE)

Using load convention on all bipoles (even sources), the sum of all powers is 0:

$$\sum p_i = 0 \iff \sum p_{\text{abs}} = \sum p_{\text{sup}}$$

Proof.

By recalling the definition of power: $\sum p_k = \sum v_k i_k$

By recalling then the definition of potential we can arrive to: $\frac{1}{2} \sum_i^n \sum_j^n (U_i - U_j) i_{ij} \forall i \neq j$

We are doing this sum on all branches (i, j are the two nodes), this means we are considering every node, thus the division by 2 is necessary

by distributing and permuting we obtain: $\frac{1}{2} (\sum_i^n U_i \sum_j^n i_{ij} - \sum_j^n U_j \sum_i^n i_{ij})$

keep in mind that for every branch (i.e. when $i = j$ in the first sum of the pairs) the i_j sum represents all entering currents, while the other are the exiting ones, thus by KCL they equal to 0, this is true for

every pair, thus:

$$\sum p_k = \frac{1}{2} \left(\sum_i^n U_i \sum_j^n i_{ij} - \sum_j^n U_j \sum_i^n i_{ij} \right) = 0$$

□

From here we can also define the non amplification theorem

THEOREM 2 (NON AMPLIFICATION THEOREM)

The theorem works both for voltage and current:

At any time the module of the current/voltage supplied by all sources is greater or equal than the module of the current/voltage on any other bipole

Proof.

For the voltage part:

Suppose the source between nodes A, B and the passive bipoles between nodes M, N_k. By KCL

$$\sum i_k = i_{MN_1} + i_{MN_2} + \dots + i_{MN_n} = 0$$

this means that there must be at least one negative current. by chosing $i_{MN_r} > 0$, $i_{MN_s} < 0$ we have

$v_{MN_r} = v_M - v_{N_r}$, $v_{MN_s} = v_M - v_{N_s}$ this means that $v_{N_r} < v_M < v_{N_s}$ and is intermediate between the two connecting ones

□

AC Regime:

It gets a bit more complicated in AC regime, since

$$p(t) = v(t)i(t) = V_M \sin(\omega t + \alpha) I_M \sin(\omega t + \beta) = V_M \sin(\omega t + \alpha) I_M \sin(\omega t + \alpha - \phi)$$

Recall that: $\sin(x) \sin(y) = \frac{1}{2} (\cos(x - y) - \cos(x + y))$

$$p(t) = \frac{V_M I_M}{2} (\cos(\phi) - \cos(2\omega t + 2\alpha - \phi)) = VI \cos(\phi) - VI \cos(2\omega t - \phi)$$

Where the first term, called **active/real power** is constant, and the second, called **fluctuating power** has pulsation 2ω .

THEOREM 3 (MEAN POWER THEOREM)

In AC regime, the mean power over one period T corresponds to the real power.

Proof.

The proof is easy and consists in a simple integration: $P = 1/T \int_{[T]} p(t) dt$

Since the fluctuating power is sinusoidal it has 0 mean, on the other hand the real power is constant.

□

By rewriting $p(t)$ it is possible to find another meaningful data:

$$p(t) = VI [\cos \phi - (\cos 2\omega t \cos \phi + \sin 2\omega t \sin \phi)] = VI \cos \phi (1 - \cos 2\omega t) - VI \sin \phi \sin 2\omega t$$

Define the **reactive power** as $Q = VI \sin(\phi)$, it corresponds to the maximum power of exchange of energy, while P is just dissipated/supplied

From P and Q we can define the **apparent power** $S = VI = \sqrt{P^2 + Q^2}$

This is just the modulus of the **complex power** $\dot{S} = \dot{V} \dot{I}^*$

This highlights how P, Q are the real/imaginary parts of the complex power

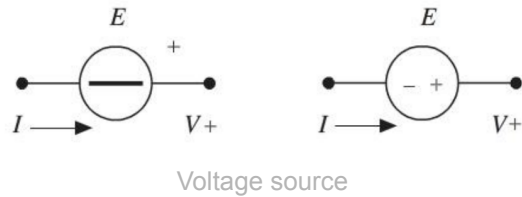
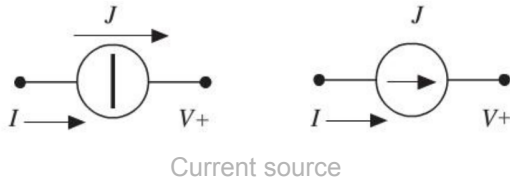
1.5) Bipoles

1.5.1) Voltage/Current Sources

Ideal:

Both these sources use the source convention, we have:

The voltage source is a bipole that keeps a constant voltage: $V_{AB} = E$. If $E = 0$ we call this a short circuit. They can't be put in parallel if they generate different voltages



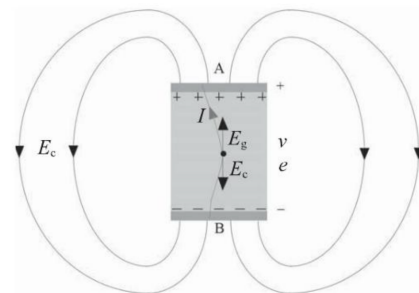
The current source is a bipole that keeps a constant current: $I = J$. If $J = 0$ we call this an open circuit. They can't be put in series if they generate different currents.

How is a generator made?

A generator doesn't create energy, it converts another type of energy (e.g. chemical) into electric by creating a difference of potential. The field separating the charges is called E_g and goes $- \rightarrow +$ inside the generator. This produces a difference of potential in the two terminals. The integral of this field is called fem (electromotive force) $\int_{-}^{+} E_g dt$.

By looking at the picture it is clear that this field isn't conservative since

$\oint E_g = \int_B^A E_g dt = \int_B^A -E_c dt = v_0 \neq 0$. On the other hand the columbian field is conservative since $\oint E_c = \int_A^B E_c dt = \int_B^A -E_c dt = v_0 - v_0 = 0$

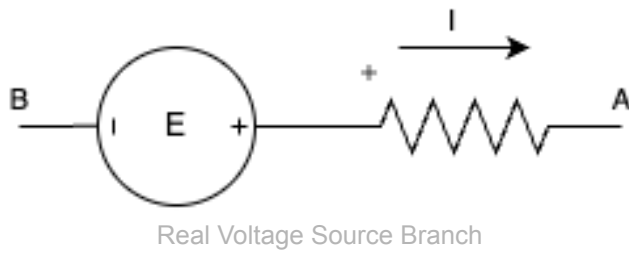


Generator with paths and directions

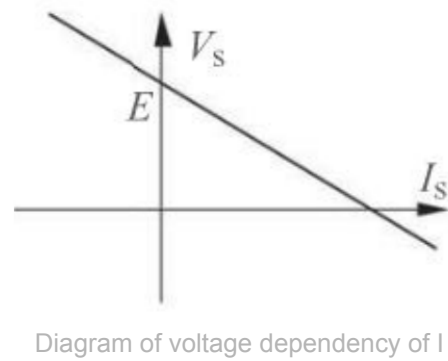
Real:

In real voltage sources the voltage delivered isn't constant, but it depends on the current given from dissipative phenomena. This voltage drop can be represented by an internal resistance put in series with the an ideal generator.

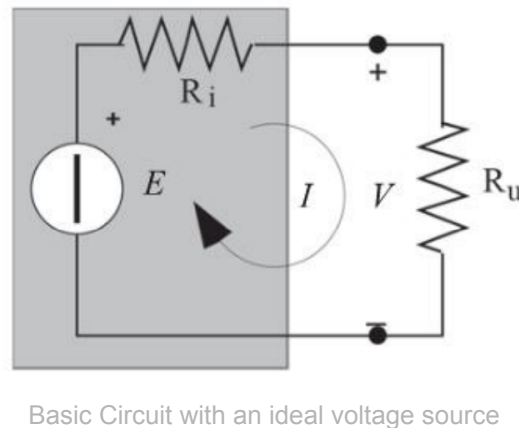
In a branch with a generator



From this image it is clear that
 $V_{AB} = -V_R + E = -RI + E$



It is clear that with 0 current the voltage generator acts as an open circuit, while holding the ideal voltage, while with $I = I_{cc} = \frac{E-V}{I}$ it acts as a short circuit.



For a **current source** we have a similar discourse, but with a parallel connection with the following characteristic equations:

$$I = J - I_R = J - \frac{V}{R_i}$$

$$V = R_i(J - I)$$

Now we will show the **performance** of a real voltage source, and it's theoretical maximum:
 In this circuit we can easily find:

$$I = \frac{E}{R_i + R_u}, V = R_u I = E \frac{R_u}{R_i + R_u}$$

Let's define the supplied and absorbed power of the generator (and R_i):

$$P_g = VI = EI = \frac{E^2}{R_i + R_u}$$

$$P_d = VI = R_i I^2 = R_i \frac{E^2}{(R_i + R_u)^2}$$

then the supplied power is: $P = P_g - P_d = R_u \frac{E^2}{(R_i + R_u)^2}$

Which is the same as the direct calculation:

$$P_g = VI = R_u I^2 = R_u \frac{E^2}{(R_i + R_u)^2}$$

We define the performance as: $\eta = \frac{P}{P_g} = \frac{R_u}{R_i + R_u}$

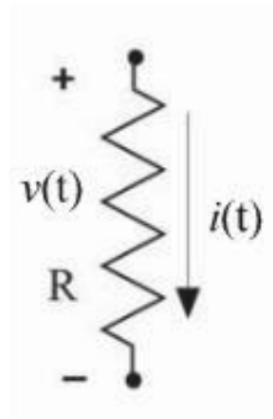
The maximum value of P can be found by derivating with respect to R_u :

$$\frac{dP}{dR_u} = E^2 \frac{R_i - R_u}{(R_i + R_u)^3}$$

Thus the max value for R_i is R_u , then, $\eta = 0.5$

1.5.2) Resistor

DC Regime:



Resistor bipole

For a linear resistor we use the **Load**

Convention. The characteristic equation is:

$v_{AB}(t) = R i(t)$ given from Ohm's law

The inverse of a resistance is the **Conductance**

$G = 1/R$ and then: $i(t) = G v_{AB}(t)$

For a resistor the power corresponds to Joule's

law: $P = -|V_{AB}I| = RI^2$ and **it is always**

negative since it is a passive bipole using the load convention

THEOREM 4 (SERIES/PARALLEL CONNECTION OF RESISTORS)

The series connection is: $R_{eq} = \sum R_i$

Proof.

In a series connection The current going through all the resistors is the same:

$$V_i = R_i I, \text{ then: } V_{AB} = \sum V_i = \sum R_i I = I \sum R_i \implies R_{eq} = \frac{V}{I} = \sum R_i$$

The parallel connection is: $\frac{1}{R_{eq}} = \sum \frac{1}{R_i}, R_{eq} = (\sum \frac{1}{R_i})^{-1}$

Proof.

In a parallele connection the voltage at the terminals is the same:

$$V = R_i I_i, \text{ then: } I = \sum I_i = V \sum \frac{1}{R_i} \implies R_{eq}^{-1} = \frac{I}{V} = \sum \frac{1}{R_i}$$

COROLLARY 5 (VOLTAGE/CURRENT DIVIDER)

In a series connection, the voltage of a single resistor can be found using the formula:

$$V_i = R_i I_i = R_i I = V \frac{R_i}{\sum R_n}$$

In a parallel connection, the current on a single resistor is:

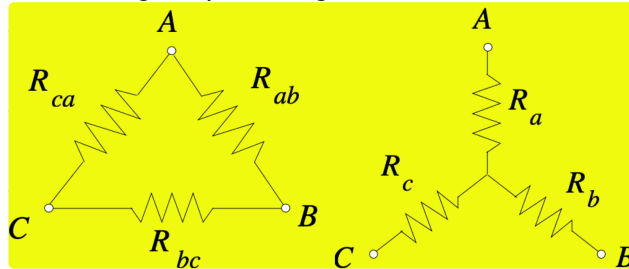
$$I_i = G_i V = G_i \frac{I}{G_{eq}} = I_i \frac{G_i}{\sum G_n}$$

From here the case with 2 resistors is obvious:

$$I_a = I \frac{R_b}{R_a + R_b}$$

THEOREM 6 (STAR/TRIANGLE TRANSFORMATION)

It is possible to transform the following shapes using these relations



Triangle and Star

$$R_{ab} = R_a + R_b + \frac{R_a R_b}{R_c} \iff R_a = \frac{R_{ab} R_{ac}}{\sum R_{ij}}$$

In AC it is interesting to study the resistive impedance:

$$Z_R = X_R = R$$

It is also important to notice that in a resistance, both current and voltage are in phase.

Proof:

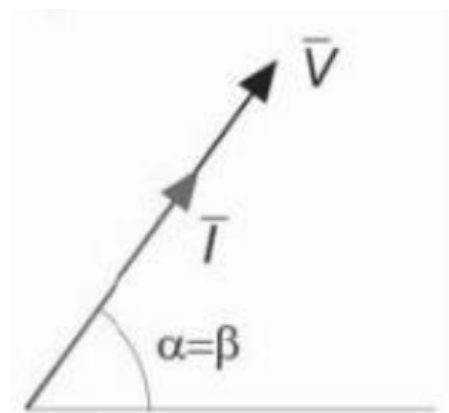
By using the definition of voltage and current:

$$v(t) = V_M \sin(\omega t + \alpha) \rightarrow i(t) = \frac{v(t)}{R} = \frac{V_M}{R} \sin(\omega t + \alpha)$$

In symbolic representation:

$$\dot{I} = \frac{\dot{V}}{R} \implies \dot{Z} = \frac{\dot{V}}{\dot{I}} = R$$

□



V and I phasors in phase

It is also useful to study the power:

Since $\phi = 0$, we have only active power: $\dot{S} = \dot{V}\bar{I} = VI = P$

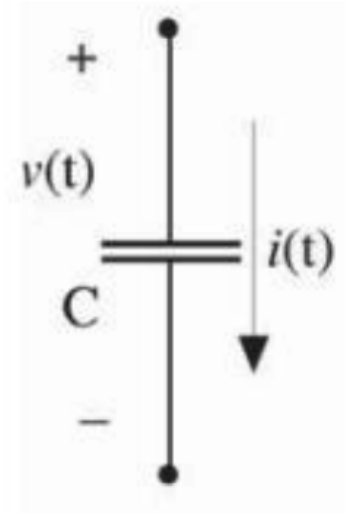
1.5.3) Capacitor

A capacitor is made of two plates of conducting material put near each other. These plates get charged $\pm Q$ and create an electric field inside the bipole proportional to Q in equilibrium conditions.

We define a constant called **capacitance** $C = \frac{Q}{v}$ measured in farad [F] positive and constant in time since it is shape dependant.

Being a bipole, the current entering must be the same as the exiting one:

$$i_a(t) = i_b(t) \implies q_a(t) + q_b(t) = 0.$$



Capacitor bipole

The bipole uses the **load convention**. Let's define it's characteristic function:

By definition of current $i = \frac{dq}{dt}$ and capacitance $C = \frac{Q}{v}$ we can find $i(t) = \frac{d(Cv(t))}{dt}$

$$i(t) = C \frac{dv(t)}{dt}$$
$$v(t) = v(0) + \frac{1}{C} \int_0^t i(\tau) d\tau$$

DC Regime:

It is important to notice that in DC regime $v(t)$ is constant, thus $i(t) = 0$ and C behaves as an open circuit.

AC Regime

In AC it is interesting to study the capacitive impedance:

$$Z_C = jX_C = j \frac{-1}{\omega C}$$

It is also important to notice that in a capacitor, current and voltage are in quadrature anticipation .

Proof:

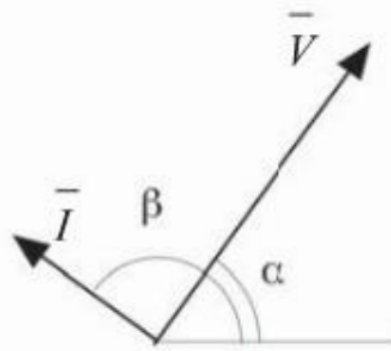
By using the definition of voltage and current:

$$v(t) = V_M \sin(\omega t + \alpha) \rightarrow i(t) = C \frac{dv(t)}{dt} = \omega C V_M \cos(\omega t + \alpha) = \omega C V_M \sin(\omega t + \alpha + \pi/2)$$

from here we get $I = \omega C V_M$. In symbolic representation:

$$\dot{I} = \omega C \dot{V} e^{j\pi/2} = j\omega C \dot{V} \implies \dot{Z} = \frac{\dot{V}}{\dot{I}} = \frac{1}{j\omega C} = \frac{-j}{\omega C}$$

□



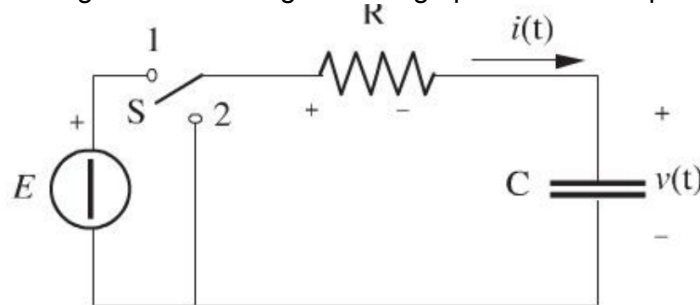
Quadrature delay

It is also useful to study the power:

Since $\phi = \pi/2$, we have only reactive power: $\dot{S} = \dot{V}\bar{I} = \omega CV^2 e^{-j\phi} = -j\omega CV^2 = -j\frac{V^2}{X_C} = -jVI = Q_C$

Variable regime

An important study in variable regime is the charge/discharge process of a capacitor:



Circuit for charge/discharge of capacitor

- Charge process:**

The switch is set in position 1 with initial condition $v_c(\pm 0) = v_c(0) = 0$.

In this case the expressions are:

$$v_C(t) = E(1 - e^{-t/RC})$$

$$i_C(t) = C \frac{dv(t)}{dt} = \frac{E}{R} e^{-\frac{t}{RC}}$$

CALCULATIONS

By Kirchhoff and the definition of capacity we have:

$$\begin{cases} i_C(t) = i_R(t) = i_E(t) \\ E = v_C + v_R \\ i_C(t) = C \frac{dv_C(t)}{dt} \end{cases} \implies E = v_C + RC \frac{dv_C(t)}{dt}$$

The forced solution, having an open circuit is: $v_{Cf} = E$

The free evolution is given by: $RCs + 1 = 0 \implies s = -\frac{1}{RC} \implies v_{Co} = Ae^{-t/RC}$

Then we have: $v_C(t) = E + Ae^{-t/RC}$

To find A we impose the initial condition: $v_c(0) = 0 = E - A \implies A = -E$

Finally we have: $v_C(t) = E(1 - e^{-t/RC})$

And $i_C(t)$ can be found by derivation

We can find the **stored energy in a capacitor**:

$$E = \int_0^\infty v(t)i(t)dt = \int_0^Q v(t)dq = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} CV^2$$

- **Discharge process**

The switch is then set in position 2 with initial condition $v_c(\pm 0) = v_c(0) = E$

$$v_C(t) = Ee^{-t/RC}$$

$$i_C(t) = -\frac{E}{R}e^{-t/RC}$$

 CALCULATIONS

Now we can set:

$$\begin{cases} i_C(t) = i_R(t) \\ 0 = v_C + v_R \\ i_C(t) = C \frac{dv_C(t)}{dt} \end{cases} \implies 0 = v_C + RC \frac{dv_C(t)}{dt}$$

The forced solution is null, while the free solution is the same as before, then we have:

$$v_C(t) = Ae^{-t/RC}$$

by imposing the initial condition we can find A: $v_C(0) = E = A$

Finally we have: $v_C(t) = Ee^{-t/RC}$

Another way to look at energy is infinitesimally by setting:

$$\begin{array}{lll} dW_{\text{supp}} & = dW_{\text{acc}} & + dW_{\text{diss}} \\ \int E i dt & = \int R i^2 dt & + \int v i dt \\ EQ & = \frac{1}{2} EQ & + \frac{1}{2} EQ \end{array}$$

thus the efficiency of charge is $\eta = 0.5$.

THEOREM 7 (SERIES AND PARALLEL CONNECTION OF CAPACITORS)

The equivalent capacity is described as:

$$C_{eq} = \left(\sum \frac{1}{C_i} \right)^{-1}, \quad \text{in series}$$

$$C_{eq} = \sum C_i, \quad \text{in parallel}$$

Proof.

Proof for series connection:

Since: $i = i_i = i_j$ and $v = \sum v_i$ it clearly becomes $v = q \sum \frac{1}{C_i} \implies C_{eq} = \frac{v}{q}^{-1} = \left(\sum \frac{1}{C_i} \right)^{-1}$

Proof for parallel connection:

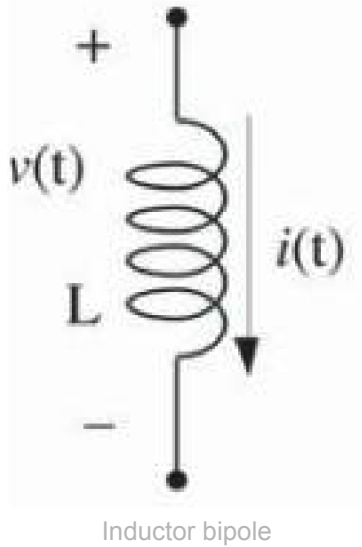
Since: $i = \sum i_i$ and $v = v_i = v_j$ it clearly becomes $q = \sum q_i = \sum C_i v_i = v \sum C_i \implies C_{eq} = q/v = \sum C_i$

□

1.5.4) Inductor

An inductor works in such a way that it inhibits current changes in a circuit due to magnetic induction.

The bipole uses the **load convention** and in order for it to be a bipole it must hold the



existence conditions: the current passing is constant and since *externally of L the induction is null* the bipole has a difference of potential on it's terminals. Here are the characteristic equations:

$$v = L \frac{di}{dt}$$

$$i(t) = i(t_0) + \frac{1}{L} \int_{t_0}^t v dt$$

By recalling Lorentz law: $Li = \Phi_B$, by derivating $d\Phi_B/dt = L di/dt$ and by Faradays law: $d\Phi_B/dt = v$

DC Regime:

It is important to notice that in DC regime $i(t)$ is constant, thus $v(t) = 0$ and L behaves as a closed circuit.

AC Regime

In AC regime it is interesting to study the inductive impedance:

$$Z_L = jX_L = \omega L$$

It is also important to notice that in an inductor, current and voltage are in quadrature delay.

Proof:

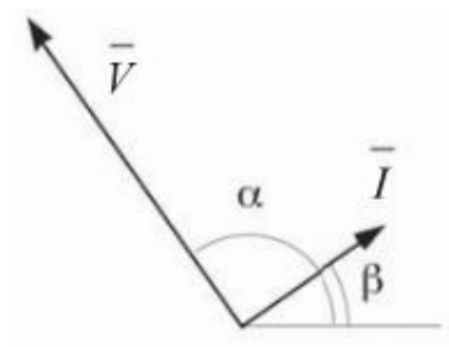
By using the definition of voltage and current:

$$v(t) = V_M \sin(\omega t + \alpha)$$

$$i(t) = \int \frac{v(t)}{L} dt = \frac{V_M}{L} \int \sin(\omega t + \alpha) = -\frac{V_M}{\omega L} \cos(\omega t + \alpha) = \frac{V_M}{\omega L} \sin(\omega t + \alpha - \pi/2)$$

from here we get $I = \frac{V_M}{\omega L}$. In symbolic representation:

$$\dot{I} = \frac{\dot{V}}{\omega L} e^{j(-\pi/2)} = -j \frac{1}{\omega L} \dot{V} \implies \dot{Z} = \frac{\dot{V}}{\dot{I}} = \frac{\omega L}{-j} = j\omega L$$



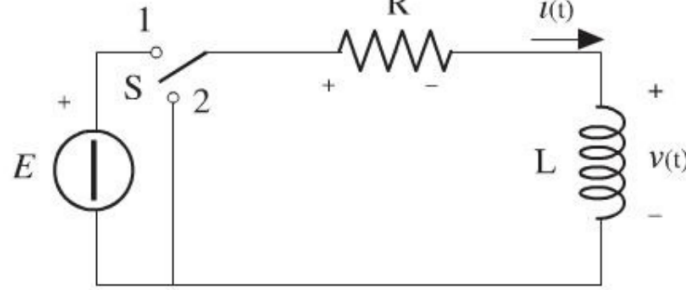
Quadrature anticipation

It is also useful to study the power:

Since $\phi = -\pi/2$, we have only reactive power: $\dot{S} = \dot{V}\bar{I} = \frac{V^2}{\omega L} e^{-j\phi} = j \frac{V^2}{X_L} = jVI = Q_L$

Variable regime

An important study in variable regime is the charge/discharge process of an inductor:



Circuit for charge/discharge of an inductor

- Charge process:**

The switch is set in position 1 with initial condition $i_L(\pm 0) = i_L(0) = 0$.

In this case the expressions are:

$$i_L(t) = \frac{E}{R}(1 - e^{-\frac{R}{L}t})$$

$$v_L(t) = L \frac{di_L(t)}{dt} = E e^{-\frac{R}{L}t}$$

CALCULATIONS

By Kirchhoff we have:

$$\begin{cases} i_L(t) = i_R(t) = i_E(t) \\ E = v_L + v_R \\ v_L(t) = L \frac{di_L(t)}{dt} \end{cases} \implies E = L \frac{di_L(t)}{dt} + Ri_L$$

The forced solution is $i_{L_f} = \frac{E}{R}$

The free solution is given by: $Ls + R = 0 \rightarrow s = -\frac{R}{L}$

then: $i_L(t) = \frac{E}{R} + Ae^{-\frac{R}{L}t}$

By imposing the initial condition: $A = -\frac{E}{R}$

- Discharge process**

The switch is set in position 2 having initial condition $i_L(\pm 0) = i_L(0) = \frac{E}{R}$

in this case the expressions are:

$$i_L(t) = \frac{E}{R} e^{-\frac{R}{L}t}$$

$$v_L(t) = L \frac{di_L(t)}{dt} = -E e^{-\frac{R}{L}t}$$

CALCULATIONS

By Kirchhoff we have:

$$\begin{cases} i_L(t) = i_R(t) \\ 0 = v_L + v_R \\ v_L(t) = L \frac{di_L(t)}{dt} \end{cases} \implies 0 = L \frac{di_L(t)}{dt} + Ri_L$$

The forced solution is $i_{L_f} = 0$

The free solution is given by: $Ls + R = 0 \rightarrow s = -\frac{R}{L}$

then: $i_L(t) = Ae^{-\frac{R}{L}t}$

By imposing the initial condition: $A = \frac{E}{R}$

Another way to look at energy is infinitesimally by setting:

$$\begin{aligned} dW_{\text{supp}} &= dW_{\text{acc}} + dW_{\text{diss}} \\ \int L i dt &= \int R i^2 dt + \int L i dt \\ L I^2 &= \frac{1}{2} L I^2 + \frac{1}{2} L I^2 \end{aligned}$$

thus the efficiency of charge is $\eta = 0.5$.

THEOREM 8 (SERIES AND PARALLEL CONNECTION OF INDUCTORS)

The equivalent inductance is:

$$\begin{aligned} L_{eq} &= \left(\sum \frac{1}{L_i} \right)^{-1}, & \text{in parallel} \\ L_{eq} &= \sum L_i, & \text{in series} \end{aligned}$$

Proof.

For series connection:

All currents are the same, moreover $v = \sum v_i \rightarrow \sum L_i i' = i' \sum L_i$ then $v/i' = \sum L_i$

For parallel connection:

All voltages are the same, moreover $i = \sum i_i \rightarrow \sum i_i(0) + (\sum 1/L_i) \int_0^t v dt$, then

$L_{eq} = (i'/v)^{-1} = (\sum 1/L_i)^{-1}$ where the last step was made by deriving i

□

1.6) Kirchhoff Laws

From the assumptions of an electrical network we can define 2 main laws called Kirchhoff laws.

- **Kirchhoff Current Law (KCL)**

The sum of the currents entering and exiting a node (or a surface cutting the network, insieme di taglio) is null:

$$\sum I_i = 0 \iff \sum I_{\text{in}} = \sum I_{\text{out}}$$

- **Kirchhoff Voltage Law (KVL)**

The sum of the voltages in a loop or mesh is 0

$$\sum V_i = 0 \iff \sum V_+ = \sum V_-$$

These two laws can be used to solve any network: define with n as the number of nodes and l the number of branches, then the following equations are needed:

- $n - 1$ KCL equations
 - $l - (n - 1)$ KVL equations
- these are all linearly independent

1.7) Theorems

THEOREM 9 (SUBSTITUTION THEOREM)

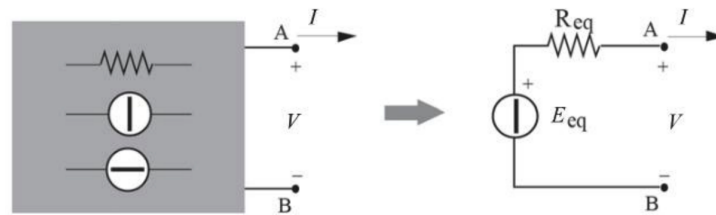
Any branch made of bipoles that admit univocal v-i relation can be rewritten as just a voltage/current source. E must be the difference of potential of the branch, J must be the current entering/exiting.

Proof.

Simply, if the branch has the same current/voltage then it can be either made of 1 bipole or 100, it doesn't change the workings of the network

THEOREM 10 (THEVENIN THEOREM: EQUIVALENT VOLTAGE GENERATOR)

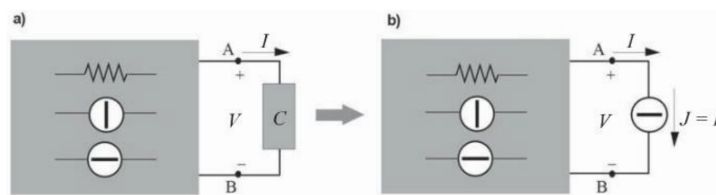
In a **linear circuit** the behaviour of any bipole network, in terms of voltage at the terminals A, B is equivalent to a real voltage source ($E_{eq} + R_{eq}$) called **equivalent voltage source**.



Thevenin theorem representation

with $E_{eq} = V_0$ the tension between the terminals A, B and R_{eq} the equivalent resistance with all generators turned off. Keep in mind $R_{eq} = \frac{V_0}{I_{cc}}$

Proof.



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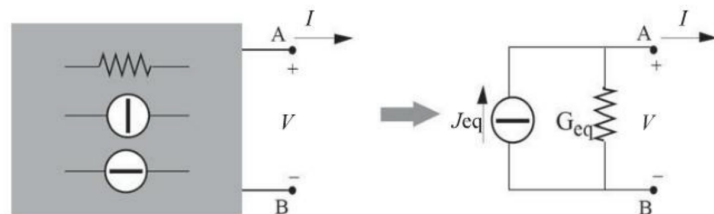
Any bipole network on the right side can be substituted by an ideal current source by substitution theorem. Then, by superimposition principle $V = V' + V''$.

V' is the voltage with J switched off. This coincides with V_0 since J acts as an open circuit

V'' is the voltage with all the generators on the left side switched off. This becomes: $V'' = -JR_{eq}$ then we have: $V = V_0 - JR_{eq}$ which is the characteristic equation of a branch with a real voltage source

THEOREM 11 (NORTON THEOREM: EQUIVALENT CURRENT SOURCE)

In a **linear circuit** the behaviour of any bipole network, in terms of voltage at the terminals A, B is equivalent to a real current source ($J_{eq} // R_{eq}$) called **equivalent current source**.



Norton Theorem representation

with $J_{eq} = I_{cc}$ the tension between the terminals A, B and R_{eq} the equivalent resistance with all generators turned off. Keep in mind $R_{eq} = \frac{V_0}{I_{cc}}$

Proof.

Totally analogous as before.

Any bipole network on the right side can be substituted by an ideal voltage source by substitution theorem. Then, by superimposition principle $V = V' + V''$.

V' is the voltage with E switched off. This coincides with V_0 since E acts as a closed circuit

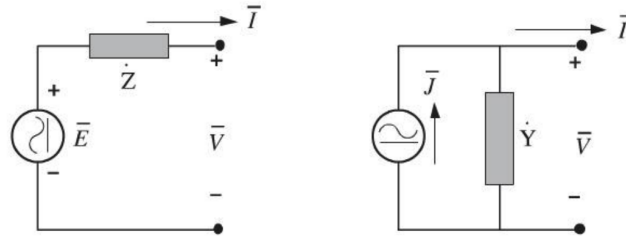
V'' is the voltage with all the generators on the left side switched off. This becomes: $V'' = -R_{eq}I_{cc}$ then we have: $V = V_0 - R_{eq}I_{cc}$ which is the characteristic equation of a branch with a real voltage

source.

□

THEOREM 12 (EQUIVALENT SYMBOLIC CURRENT / VOLTAGE SOURCE)

The behaviour of any bipole network in AC regime can be written as either an equivalent voltage source and impedance, or an equivalent current source and an admittance.



Representation of the theorem

Proof.

a

□

1.8) DC Regime

The DC regime is defined as having sources that generate their parameter constantly from infinite time for infinite time. In this regime capacitors act as open circuits and inductors as closed circuits, since they are in phase quadrature

1.9) AC Regime

In a periodic regime the time dependent quantities follow the following law: $a(t) = a(t + nT)$. We can then define the following values:

- Max value: $A_M = \max\{a(t)\}$
- Mean over one period: $A_m = \frac{1}{T} \int_0^T a(t) dt$
- Root Mean Square (RMS) : $A = \sqrt{\frac{1}{T} \int_0^T a^2(t) dt}$

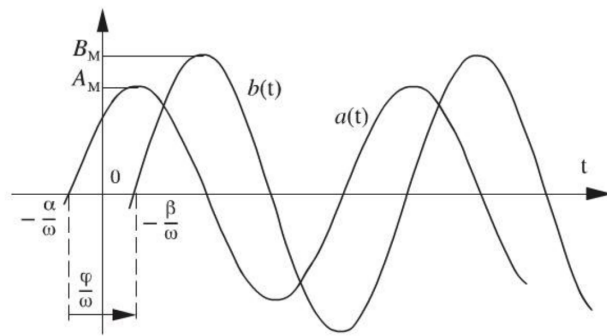
In order to be AC the mean $A_m = 0$, and the period can be identified by two symmetric halves.

A sinusoidal function is the main function we study in this regime: $a(t) = A_M \sin(\omega t + \alpha)$, where the parameters have the following relation: $\omega = 2\pi f$. Now we can calculate the values:

- Max value: A_M
- Mean over one period: $A_m = 0$
- Mean over a half period: $A'_m = \frac{1}{T/2} \int_0^{T/2} A_M \sin(\omega t) dt = \frac{2A_M}{T} [-\frac{1}{\omega} \cos(\omega t)]_0^{T/2} = \frac{2}{\pi} A_M$
- Root Mean Square (RMS) : $A = \sqrt{\frac{1}{T} \int_0^T a^2(t) dt} = \frac{A_M}{\sqrt{2}}$
- Form factor: $K_f = \frac{A}{A'_m} = \frac{\pi}{2\sqrt{2}} = 1,11$

It is important to note that in symbolic representation the RMS is used, so $A = \frac{A_M}{\sqrt{2}}$

In the case of **2 isofrequency functions**, i.e. $a(t), b(t)$ with same frequency but different amplitude and phase the following relations might happen:



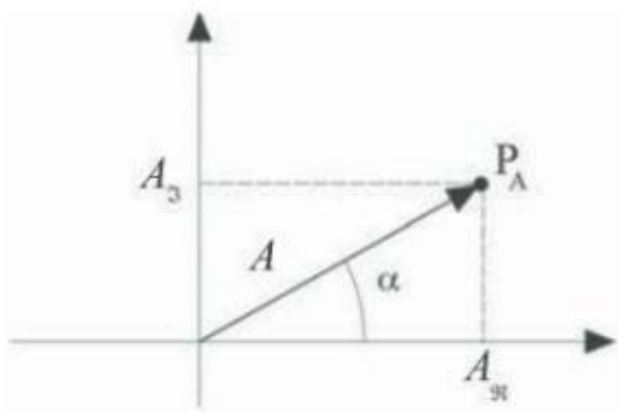
Two isofrequency functions

- In phase: $\alpha = \beta$
- Phase Opposition: $\alpha - \beta = \pi$
- Phase Anticipation: $\alpha > \beta$, in quadrature if $\alpha - \beta = \pi/2$
- Phase Delay: $\beta > \alpha$, in quadrature if $\beta - \alpha = -\pi/2$

REMARK 13 (ARCTANGENT ANGLE ERROR)

When calculating the phase of the sinusoidal representation starting from the complex representation it is important to check the sign of the real part. **If the real part is smaller than 0 it is necessary to add π to the resulting phase**

1.9.1) Phasor Representation



Graphical representation of a phasor

The phasorial representation shows $a(t)$ as a vector of modulus A_M rotating counterclockwise with angular velocity ω and starting angle α . The projection of the phasor on the x and y axis show respectively the cosine and sine function.

1.9.2) Symbolic representation

More useful is the symbolic representation. Usually a sinusoidal is expressed in the form:

$$\dot{A} = \sqrt{2}A_{\text{eff}}e^{j\alpha}e^{j\omega t} = A_M e^{j\alpha}e^{j\omega t} = A_M e^{j(\omega t + \alpha)}$$

A simple proof consists of taking the phasorial representation in complex terms:

$$\dot{A} = A_M \cos(\omega t + \alpha) + jA_M \sin(\omega t + \alpha).$$

By using:

$$\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b), \sin(a+b) = \sin(a)\cos(b) + \sin(b)\cos(a)$$

we get:

$$\begin{aligned}
A_M(\cos(\omega t)\cos(\alpha) - \sin(\omega t)\sin(\alpha) + j\sin(\omega t)\cos(\alpha) + j\cos(\omega t)\sin(\alpha)) &= \\
A_M(\cos(\alpha)(\cos(\omega t) + j\sin(\omega t)) + j\sin(\alpha)(\cos(\omega t) + j\sin(\omega t))) &= \\
A_M(\cos(\alpha) + j\sin(\alpha))(\cos(\omega t) + j\sin(\omega t)) &=
\end{aligned}$$

with the term $A_M(\cos(\alpha) + j\sin(\alpha))$ being $\dot{A}$ at $t = 0$.

1.9.3) Impedance

Define a passive bipole using load convention(resistor, capacitor, inductance) in a AC circuit, we define the impedance as :

$$\dot{Z} = \frac{\dot{V}}{\dot{I}} = \frac{V e^{j\alpha} e^{j\omega}}{I e^{j\beta} e^{j\omega}} = Z e^{j(\alpha-\beta)} = Z e^{j\phi}$$

This complex operator shows the tendency of the bipole to resist the change of current passing through it. In particular, the term ϕ shows the phase difference (delay) of $\dot{I}$ with respect to $\dot{V}$.

The admittance is the reciprocal of the inductance

It is possible to define a parallel and series connection of all passive bipoles and find their equivalent impedance:

$$\begin{aligned}
\dot{Z}_{\text{series}} &= R_{\text{series}} + jX_{\text{series}} = \sum \dot{Z}_i \\
\dot{Z}_{\text{parallel}} &= (R_{\text{parallel}} + jX_{\text{parallel}})^{-1} = (\sum 1/\dot{Z}_i)^{-1}
\end{aligned}$$

Here is a quick recap of impedance for the passive bipoles we are studying:

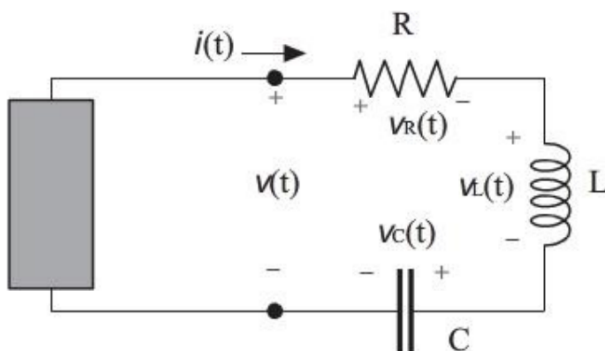
	Z	X	phase with $\bar{V}$
R	R		$\phi = 0$: in phase
C	jX_C	$-1/\omega C$	$\phi = \pi/2$: anticipation quadrature
L	jX_L	ωL	$\phi = -\pi/2$: delay quadrature

1.9.4) Resonance

A series/parallel connection of an inductor and a capacitor bipole can cause some strange behaviours. An easy way to check if two bipoles are in resonance is to see if the fundamental pulsation is given by:

$$\omega = \frac{1}{\sqrt{LC}}$$

Series resonance:



Circuit for series resonance

In such a circuit, if $\omega_0 = \frac{1}{\sqrt{LC}}$, then the inductor-capacitor series acts as a closed circuit since $\dot{Z}$ has no imaginary part:

$$\begin{aligned} \text{Im}(\dot{Z}) &= \omega L - \frac{1}{\omega C} = \frac{\omega^2 LC - 1}{\omega C} \\ &= \frac{\frac{1}{LC} LC - 1}{\omega C} = 0 \end{aligned}$$

The **resonance pulsation** was found by finding the min value of:

$$|\dot{Z}(\omega)| = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}$$

Instead of doing the derivative it is sufficient to notice that the minimum value is when $X_L = -X_C \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$ and thus current and voltage are in phase

By studying other pulsations we can come to other conclusions:

- $\omega < \omega_0$:

$$\lim_{\omega \rightarrow 0} \dot{Z}(w) = \infty \rightarrow \lim_{\omega \rightarrow 0} \phi(w) = -\pi/2$$

this means that the capacitive impedance prevails on the inductive one.

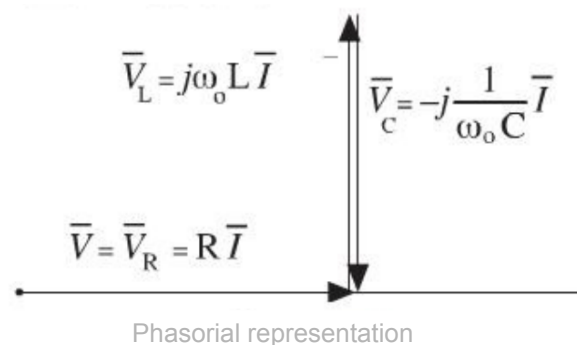
- $\omega > \omega_0$:

$$\lim_{\omega \rightarrow \infty} \dot{Z}(w) = \infty \rightarrow \lim_{\omega \rightarrow \infty} \phi(w) = \pi/2$$

this means that the inductive impedance prevails on the capacitive one.

By doing a phasorial study of the voltage, the short circuit is evident: the differences of potential in the two bipoles are equal and opposite. Moreover their Reactive power is null:

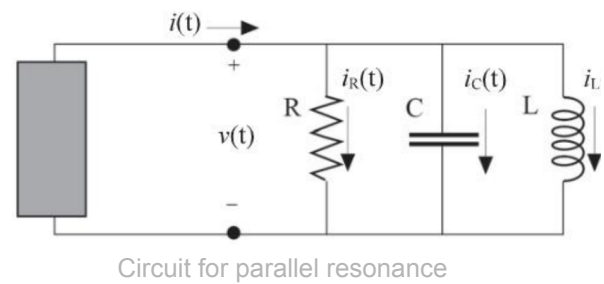
$$Q_L + Q_C = (X_L + X_C)I^2 = 0$$



Parallel resonance

In such a circuit, if $\omega_0 = \frac{1}{\sqrt{LC}}$, then the inductor-capacitor parallel acts as an open circuit since $\dot{Y}$ has no imaginary part:

$$\begin{aligned} \text{Im}(\dot{Y}) &= \omega C - \frac{1}{\omega L} = \frac{\omega^2 LC - 1}{\omega L} \\ &= \frac{\frac{1}{LC} LC - 1}{\omega L} = 0 \end{aligned}$$



The reasoning is analogous with the series resonance. Let's study what impedance prevails in with other pulses:

- $\omega < \omega_0$:

$$\lim_{\omega \rightarrow 0} \dot{Y}(w) = \infty \rightarrow \lim_{\omega \rightarrow 0} \phi(w) = \pi/2$$

this means that the inductive impedance prevails on the capacitive one.

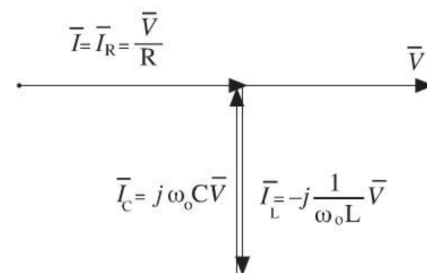
- $\omega > \omega_0$:

$$\lim_{\omega \rightarrow \infty} \dot{Y}(w) = \infty \rightarrow \lim_{\omega \rightarrow \infty} \phi(w) = -\pi/2$$

this means that the capacitive impedance prevails on the inductive one.

By doing a phasorial study of the current, the open circuit is evident: the currents in the two bipoles are equal and opposite. Moreover their Reactive power is null:

$$Q = \frac{V^2}{X_L + X_C} = \left(\frac{1}{\omega_0 L} - \omega_0 C \right) V^2 = 0$$



Phasorial representation

1.9.5) Periodic Regime

Any periodic function can be written as a sum of sinusoidal waves with different amplitudes and integer multiple of one fundamental pulsation (see SESI notes).

Every generator can be written as a sum of a generator with the n -th harmonic and then by superimposition we can find $V = \sqrt{\sum V_n^2}$, $I = \sqrt{\sum I_n^2}$ and $|\dot{S}| = \sqrt{\sum V_n^2 \sum I_n^2}$ and thus $S^2 \neq P^2 + Q^2$ and a higher apparent power is present in a circuit.

1.10) Variable Regime

We have studied this chapter in a less theoretical way, focusing more on the methods of resolution of an exercise.

An exercise will be divided in 2 parts divided at $t = 0$ by a topological change in the circuit.

First it is necessary to find $i_L(0^-), v_C(0^-)$ for every inductor and capacitor in the circuit with $t < 0$. These values will also hold in $t = 0^+$ after the topological change in the circuit.

Then, at $t > 0$, it is necessary to write all the KVL and KCL and find the differential equation that describe the desired $i_L(t)$ and/or $v_C(t)$.

once found it will look something like this: $a \frac{d^2v}{dt^2} + b \frac{dv}{dt} + cv = d$
the result will be of form: $v(t) = v_f(t) + v_o(t)$

- The forced solution is obtained by solving the circuit at $t > 0$ as if it were at regime
- The free solution is obtained by transforming the differential equation in: $as^2 + bs + c = 0$ and then by finding s that can be:
 - s real: $s \in \mathbb{R}$ will have solution of type Ae^{-st}
 - s imaginary complementary couple: $s_{1,2} = \sigma \pm j\omega$ will have solution of form $e^{\sigma t}(A \sin(\omega t) + B \cos(\omega t))$

To find A, B impose the initial conditions and the exercise is solved

Be careful of open/closed circuits and single branches. It often happens that a circuit must be solved as many smaller independent circuits